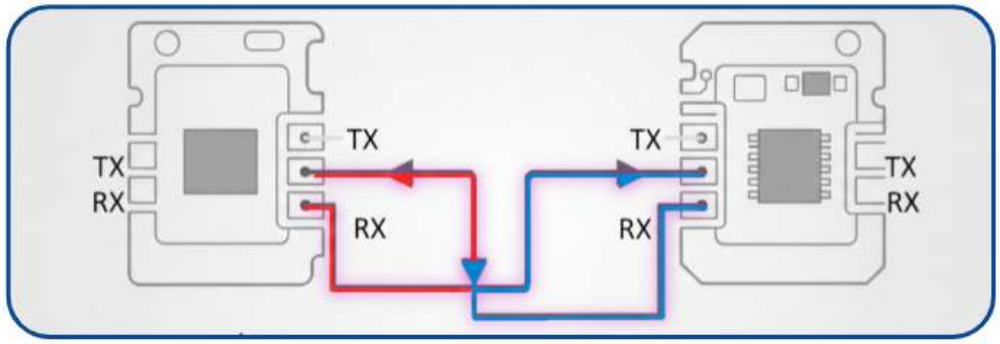
UART: The Asynchronous Workhorse

Click the numbered buttons to learn more.

- **TX (Transmit):** The line for sending data.
- **RX (Receive):** The line for receiving data.
- The TX of one device connects to the RX of the other, and vice-versa.

Key Feature: Asynchronous

- There is no shared clock signal between the transmitter and receiver.



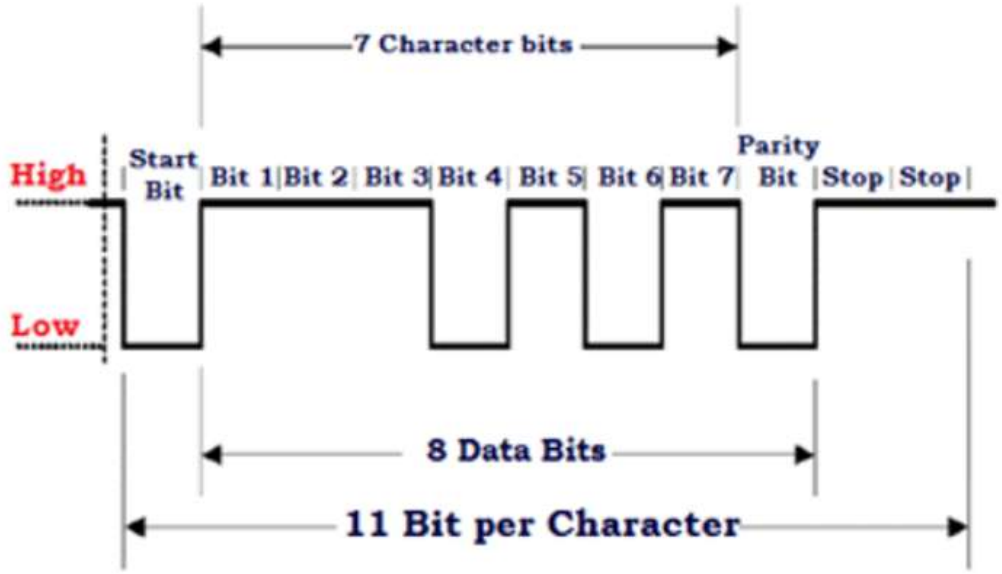
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UART Data Framing: Creating Structure Without a Clock

- **The Solution:** Both devices agree on the timing (Baud Rate) and the structure (Frame Format) beforehand.
- **The UART Frame:** A packet of bits with a defined structure.
- **Idle State:** The line is held HIGH when no data is being sent.
- **Start Bit:** A single LOW bit that signals the beginning of a frame.
- **Data Bits:** The actual data payload (typically 5 to 9 bits, LSB first).
- **Parity Bit (Optional):** Used for simple error checking (even or odd parity).
- **Stop Bit(s):** One or two HIGH bits that signal the end of the frame.



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UART Baud Rate: The Speed Agreement

What is Baud Rate?

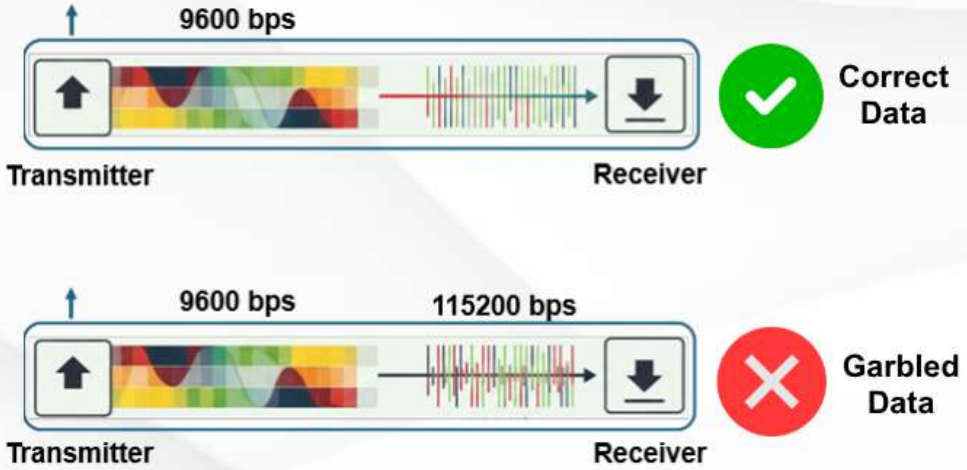
- The rate at which symbols (in this case, bits) are transferred. Measured in bits per second (bps).
- It defines the duration of each bit in the UART frame.

The Golden Rule of UART:

- Western Digital created one of the first widely available single-chip UARTs, the WD1402A, around 1971.
- Became the standard for connecting PCs to modems and printers. The National Semiconductor 8250 was famously used in the first IBM PC.

Common Baud Rates:

- 9600 bps (very common for simple devices)
- 19200 bps
- 57600 bps
- 115200 bps (common for debugging and faster transfers)



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SPI: The Synchronous Speedster

What it is:

- Serial Peripheral Interface. A synchronous serial protocol.

Key Feature: Synchronous

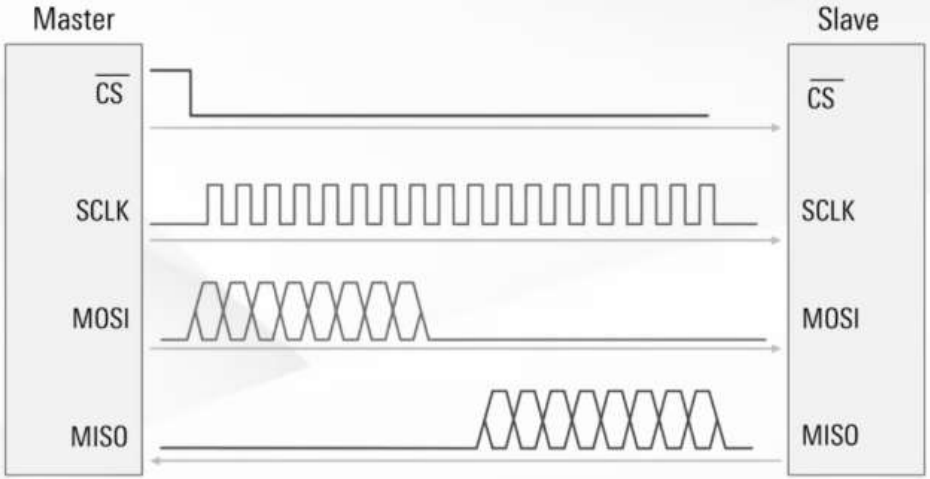
- A dedicated clock line (SCK) is controlled by the master.
- This eliminates the need for baud rate agreement and allows for much higher speeds.

Architecture: Master-Slave

- One Master device (e.g., the Renesas QuickConnect board) controls the communication.
- One or more Slave devices (sensors, memory, etc.) respond to the master.

Communication is Full-Duplex:

- Data can be sent from master to slave and slave to master simultaneously on every clock cycle.



SPI: Master-Slave Architecture

Click the numbered buttons to learn more.

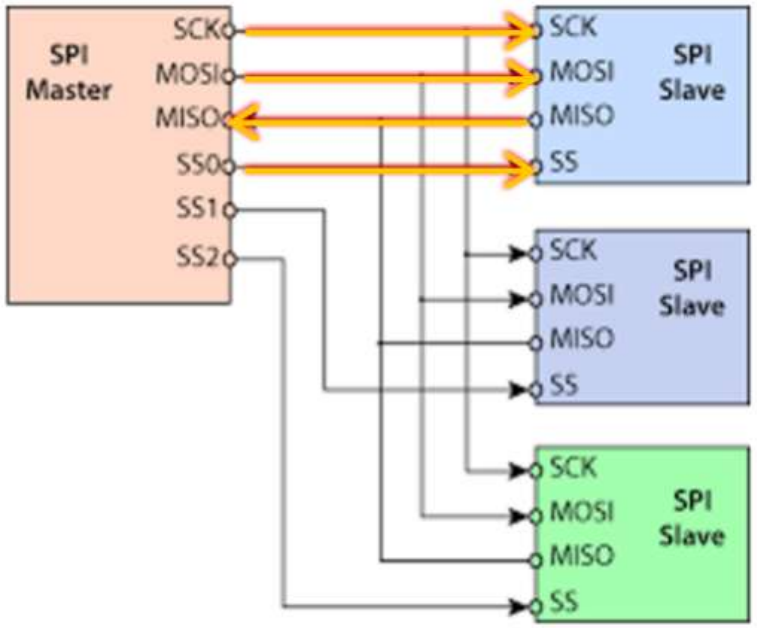
The Four SPI Lines

SCK (Serial Clock): The clock signal, driven by the Master.

MOSI (Master Out, Slave In): Data from the Master to the Slave.

MISO (Master In, Slave Out): Data from the Slave to the Master.

SS / CS (Slave Select / Chip Select): The Master uses this line to select a specific Slave. It is an active-low signal.



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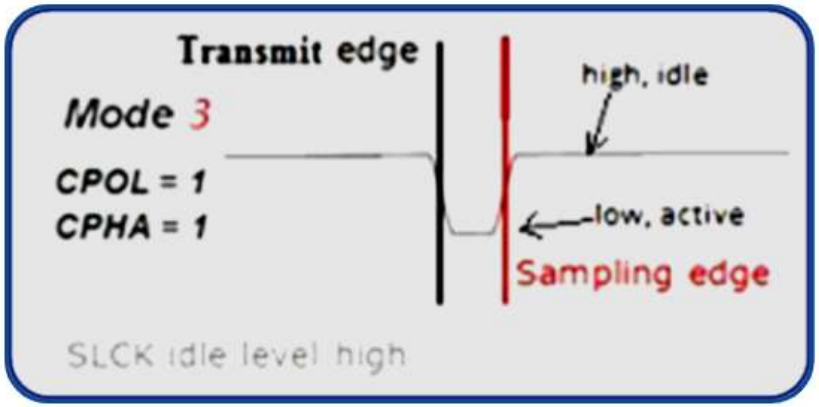
SPI Modes: Clock Polarity & Phase

Click the numbered buttons to learn more.

- **CPOL (Clock Polarity):** Defines the idle state of the clock.
- **CPOL=0:** Clock is idle LOW.
- **CPOL=1:** Clock is idle HIGH.

CPHA (Clock Phase): Defines when data is sampled.

- **CPHA=0:** Data is sampled on the leading (first) edge of the clock pulse.
- **CPHA=1:** Data is sampled on the trailing (second) edge of the clock pulse.



1 2

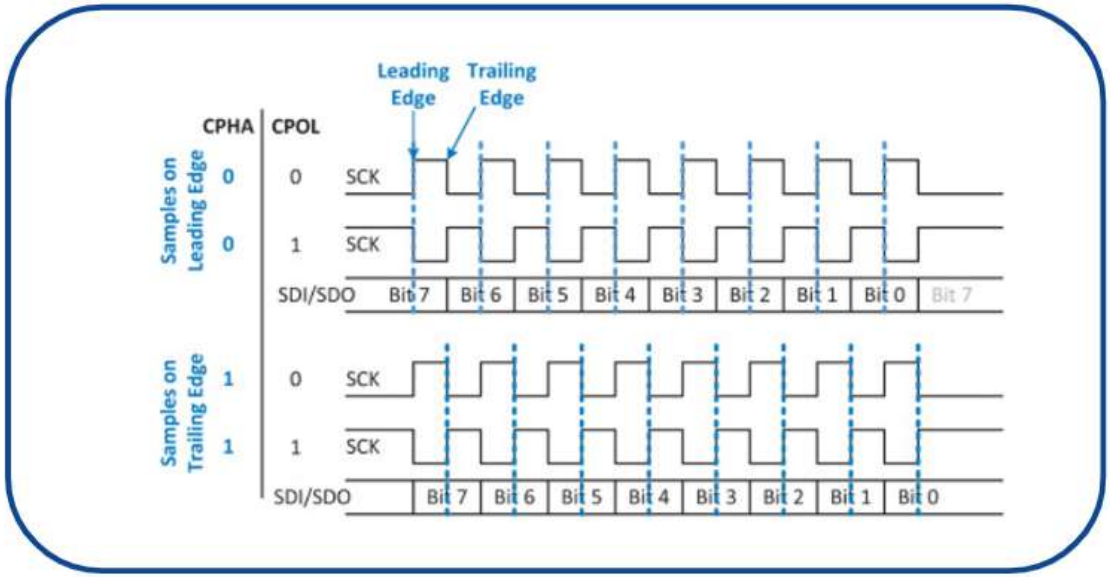
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SPI Modes: Clock Polarity & Phase

The Four SPI Modes:

- **Mode 0:** CPOL=0, CPHA=0
- **Mode 1:** CPOL=0, CPHA=1
- **Mode 2:** CPOL=1, CPHA=0
- **Mode 3:** CPOL=1, CPHA=1



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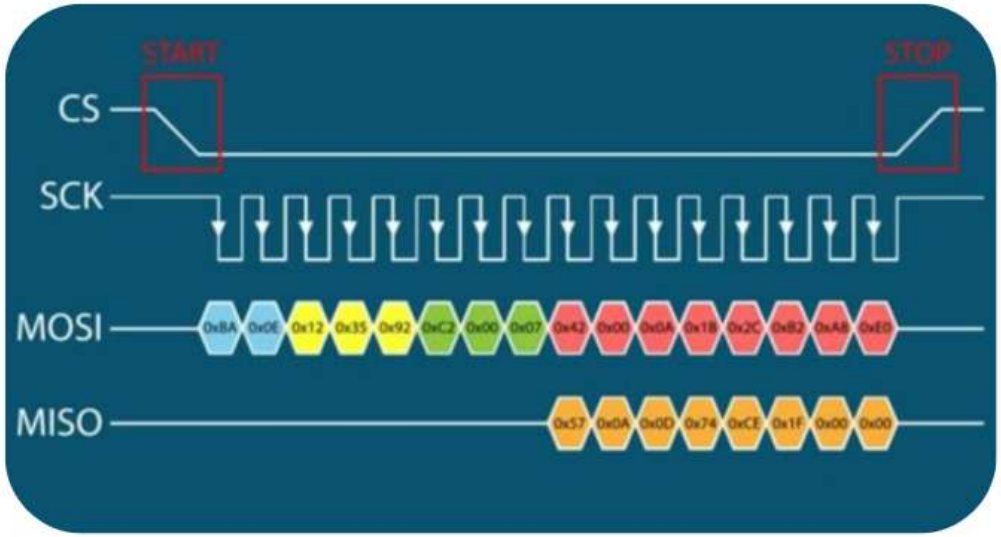
The Working of SPI

The Transaction:

1. Master pulls the desired slave's CS line LOW.
2. Master generates clock pulses on SCK.
3. On each clock pulse, the master sends a bit on MOSI and the slave sends a bit on MISO. This is full-duplex.
4. Master pulls the CS line HIGH to end the transaction.

The Agreement:

Master and slave must agree on the SPI Mode (CPOL/ CPHA).



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I2C: The Efficient Two-Wire Bus

What it is: Inter-Integrated Circuit. A synchronous, multi-master, multi-slave serial bus.

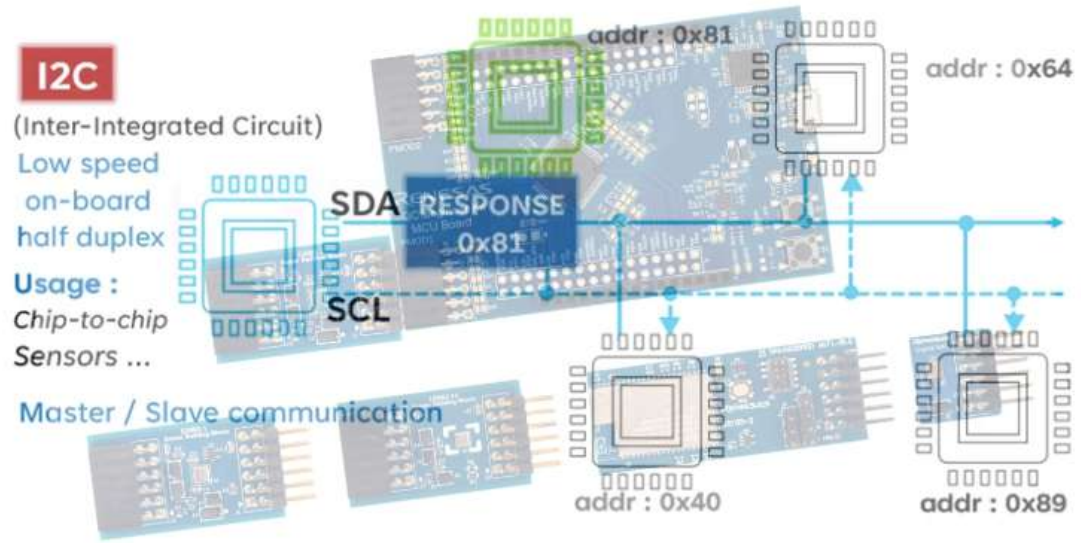
Key Advantage: Connects multiple devices using only two wires.

The Two I2C Lines:

- SCL (Serial Clock): The clock line, shared by all devices.
- SDA (Serial Data): The data line, shared by all devices.

Open-Drain Bus:

- Devices can only pull the lines LOW. External pull-up resistors hold the lines HIGH when idle.
- This prevents bus contention and is key to enabling multi-master support.



I2C: Addressing and Bus Protocol

How it Works:

- Instead of Chip Select lines, I2C uses a software addressing scheme.

The I2C Transaction:

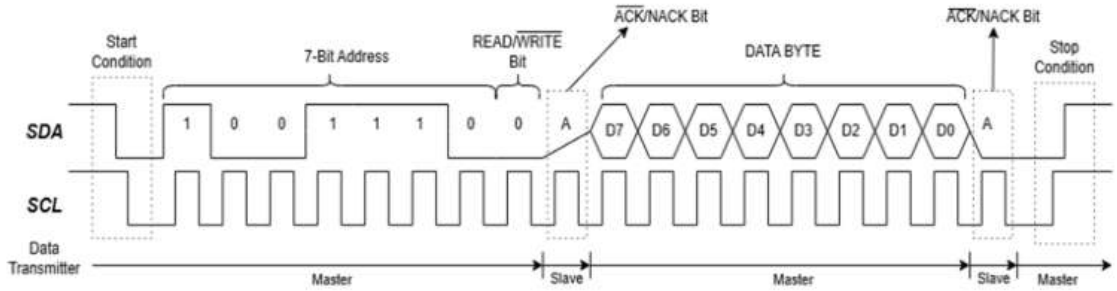
Start Condition: The Master initiates communication by pulling SDA low while SCL is high.

Address Frame: The Master sends a 7-bit slave address, followed by a Read/Write bit (0=Write, 1=Read).

ACK/NACK: The addressed Slave pulls SDA low for one clock cycle to Acknowledge (ACK) that it is ready. If it doesn't, it's a Not Acknowledge (NACK).

Data Frames: Master and Slave exchange 8-bit data bytes, with an ACK after each byte.

Stop Condition: The Master ends the communication by letting SDA go high while SCL is high.



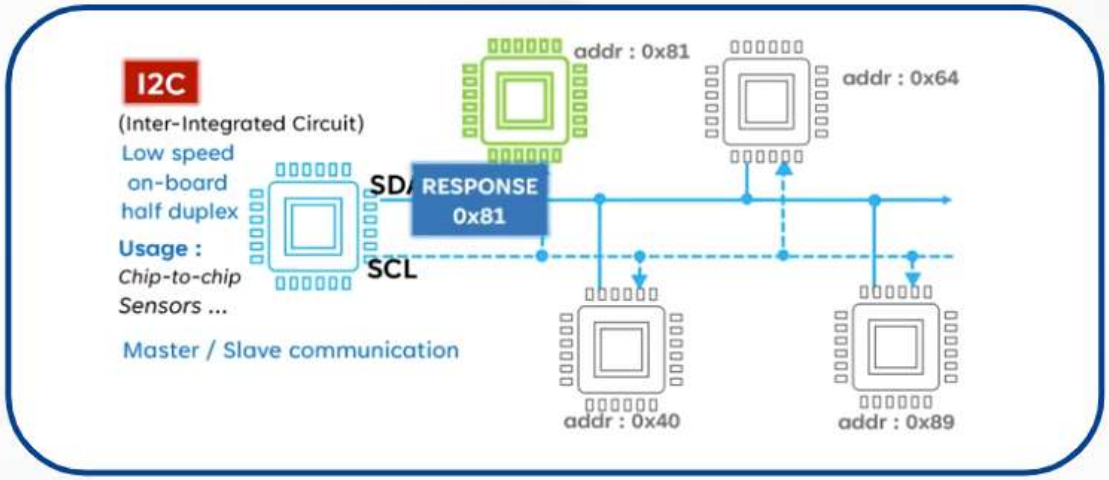
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The Working of I2C

The Transaction Flow:

1. **Start**: Master creates a unique Start Condition on the bus.
2. **Address**: Master sends the 7-bit slave address + a Read/Write bit.
3. **Acknowledge (ACK)**: The matching slave pulls SDA low to acknowledge.
4. **Data**: Master and slave exchange 8-bit data bytes, with an ACK after each byte.
5. **Stop**: Master creates a Stop Condition to release the bus.



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The Working of I2C

The Bus: A simple two-wire bus for multiple devices.

- SCL (Serial Clock): Shared clock line.
- SDA (Serial Data): Shared data line.
- Open-Drain: Requires external pull-up resistors. Devices only pull the lines LOW.

The Protocol: Software Addressing

- Each slave device has a unique 7-bit address. No chip select lines are needed.






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Protocol Showdown: UART vs. SPI vs. I2C

Feature	UART	SPI	I2C
Wires Required	2 (+GND)	4+ per slave (+GND)	2 (+GND)
Clock	Asynchronous (No Clock Line)	Synchronous (Master-driven)	Synchronous (Master-driven)
Speed	Slowest (e.g., up to 1 Mbps)	Fastest (e.g., 10+ Mbps)	Medium (100kbps, 400kbps, 1Mbps+)
Device Addressing	None (Point-to-Point only)	Hardware Chip Select line per slave	Software 7-bit/10-bit address
Duplex	Full-Duplex	Full-Duplex	Half-Duplex
Complex	Simple	Moderate	More Complex (Addressing, ACK / NAC)
Renesas Periheral	SCI	RSPI / SCI	RIIC / SCI

Putting It All Together: A Quick Connect Example

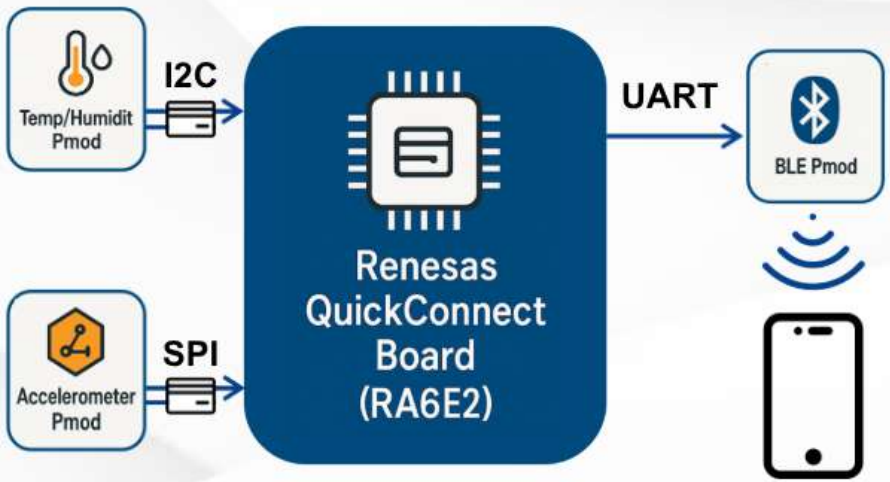
The Goal: Build a simple environmental monitor that streams data wirelessly.

The Hardware:

- **Master MCU**: Renesas RA6E2 Fast Prototyping Board (A QuickConnect IoT board).
- **I2C Sensor**: Renesas HS3001 Temperature/Humidity Pmod.
- **SPI Sensor**: A generic Pmod with an SPI-based accelerometer.
- **UART Wireless**: Renesas DA16600 BLE Pmod.

The Data Flow:

1. The RA6E2 MCU uses the I2C bus on its Pmod connector to request temperature and humidity data from the HS3001 sensor.
2. The MCU uses the SPI bus on its other Pmod connector to read motion data from the accelerometer.
3. The MCU formats the sensor data into a single data packet.
4. The MCU sends this packet over UART to the BLE Pmod, which then wirelessly transmits the data to a connected smartphone or PC.

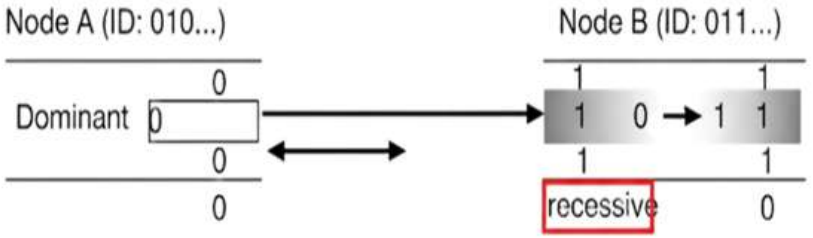


The Working of CAN: A Robust, Message-Based System

Click on the buttons to learn more.

Bus Access: Priority-Based Arbitration

- If multiple nodes start talking at once, they begin transmitting their Message IDs.
- The lower the ID number, the higher the priority.
- As nodes transmit, they also listen. If a node sends a recessive '1' but sees a dominant '0' on the bus, it knows a higher-priority message is being sent.
- The "losing" node immediately stops transmitting and becomes a listener. The highest-priority message continues without being corrupted. This is called **Non-Destructive Arbitration**.



Unparalleled Reliability: CAN Error Handling

Click the numbered buttons to learn more.

Multi-Layered Error Detection:

Every CAN controller constantly checks for errors in five different ways.

- **Bit Monitoring**: A transmitter checks that the bit it sent is the same as the bit on the bus.
- **Bit Stuffing**: After 5 identical bits, a bit of the opposite polarity is inserted. If a node sees 6 identical bits, it's a Stuff Error.
- **Frame Check**: Checks that certain parts of the message frame have the correct, fixed format.
- **Acknowledgement (ACK) Check**: The transmitter checks if at least one receiver acknowledged the message.
- **Cyclic Redundancy Check (CRC):** A 15-bit checksum ensures the data content has not been corrupted.



1 2 3

Unparalleled Reliability: CAN Error Handling

Fault Confinement: Isolating Bad Nodes

Each node keeps a Transmit and Receive Error Counter (TEC/REC).

- **Error Active**: Normal state. Can send active error frames.
- **Error Passive**: If error counts get too high, the node can no longer destroy bus traffic. It becomes a "passive“ protestor.
- **Bus Off**: If a node is clearly faulty (TEC > 255), it takes itself offline completely to protect the rest of the network.

